StandardML - Codex

Project Status Overview

XiLAB X-in-the-Loop Advanced Simulation Lab
Marcello Pellicciari marcello.pellicciari@unimore.it **Pietro Bilancia** pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Why This Project Exists

- transmission error is a key reducer accuracy signal
- the repository targets both predictive accuracy and engineering usability
- deployment realism matters, especially for future TwinCAT / PLC paths

Marcello Pellicciari marcello.pellicciari@unimore.it **Pietro Bilancia** pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Current Repository Identity

- no longer a single-model prototype
- now a structured TE modeling program
- includes training, validation, smoke-test, campaign, report, and guide layers

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

What Has Already Been Built

- validated TE dataset workflow
- shared training and verification infrastructure
- styled PDF reporting pipeline
- campaign planning and campaign-results discipline
- educational guides and NotebookLM source packages

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Implemented Model Families

- feedforward
- harmonic_regression
- periodic_mlp
- residual_harmonic_mlp
- tree
- Each family now lives inside the same artifact and evaluation structure.

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Best Current Results

Family	Best Test MAE [deg]
tree	0.002885
residual_harmonic_mlp	0.003152
feedforward	0.003301
periodic_mlp	0.003317
harmonic_regression	0.020782

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Main Technical Conclusion

- the current global leader is still hist_gradient_boosting
- the strongest neural direction is now residual_harmonic_mlp
- harmonic regression remains valuable mainly as a structured reference

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Why The Work Was Done This Way

- structured baselines before heavier models
- shared infrastructure before broad family expansion
- reporting discipline before large campaign growth
- documentation and explainability treated as core project assets

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Documentation And Communication Layer

- GitHub-facing README
- doc/README.md documentation index
- canonical guides with PDFs
- NotebookLM source packages for video-oriented reuse
- repository-owned Codex skills and subagents

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Current Objectives

- keep model comparison honest through registries and campaign reports
- preserve residual harmonic as the main neural branch
- open the TwinCAT deployment evaluation track carefully
- prepare the transition to Wave 2 temporal models

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Important Open Gaps

- neural models have not yet overtaken the tree leader
- temporal families are still pending
- deployment evaluation is planned but not yet executed
- the live backlog needs synchronization with the latest delivered reports

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Recommended Next Steps

- 1. update the live backlog and running-state view 2. formalize the TwinCAT deployment evaluation branch 3. keep the residual winner as the neural reference anchor 4. prepare Wave 2 temporal implementation 5. preserve the current documentation and reporting discipline

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia

Strategic Takeaway

- The repository has already built the foundation and the first serious comparison layer.
- The next question is no longer whether the project can run structured TE experiments.
- The next question is which direction should be promoted next without losing deployment realism, documentation quality, and comparison discipline.

Marcello Pellicciari marcello.pellicciari@unimore.it Pietro Bilancia pietro.bilancia@unimore.it
© Università degli studi di Modena e Reggio Emilia